আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম

الجامعة الإسلامية العالمية شيتاغونغ

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Computer science and Engineering (CSE)

Electronics Lab Project

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Project Information:

Project Name:	Sun Tracking Solar Panel
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Submitted to:

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Sun Tracking Solar Panel

- **Pointed features of the design:** A **sun-tracking solar panel automatically adjusts its position to follow the sun throughout the day, ensuring maximum exposure to sunlight. It uses light sensors such as LDRs to detect the direction of highest light intensity and a microcontroller to process the data. Motors rotate the panel in single-axis or dual-axis movement for precise alignment with the sun. This system significantly increases power generation compared to fixed solar panels. It includes a battery to store generated energy and power the control unit. The design is energy-efficient, reliable, and suitable for sustainable power applications in EEE projects.**

- **Hardware Requirements:**

- **Resistor:** A resistor is a passive electronic component that resists the flow of electric current, measured in ohms (Ω), used to control, limit, or divide current and voltage in circuits, protecting sensitive parts like LEDs or shaping signals, often identified by colored bands indicating value and tolerance.

Figure 1.1:Resistor

- **LDR:** LDR is a versatile acronym, most commonly meaning Long-Distance Relationship, describing romantic or friendship bonds separated by significant geography, requiring digital connection. In electronics, LDR stands for Light-Dependent Resistor, a component that changes resistance with light levels, used in sensors for automatic lights, security systems, or weather stations.

Figure 1.2: LDR

- **<u>IC (TDA2822M):</u>** The TDA2822 is a versatile, low-voltage, dual-channel audio power amplifier IC, ideal for battery-powered devices like portable radios and cassette players, supporting both stereo and Bridge-Tied Load (BTL) modes for higher power, featuring low quiescent current, thermal protection, and available in DIP or SMD packages, operating from 1.8V to 15V.

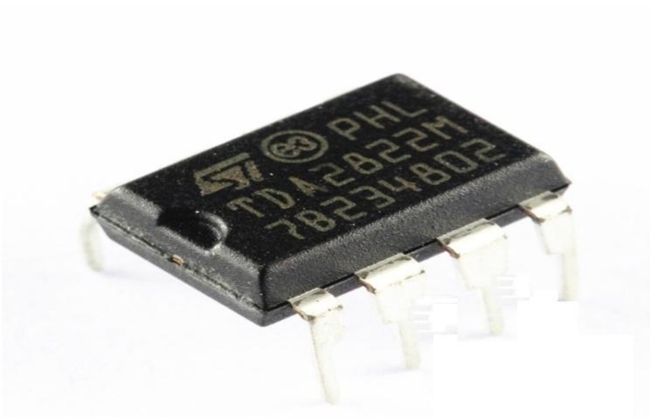

Figure 1.3: IC TDA2822M

- **<u>DC Motor:</u>**. A DC motor (Direct Current motor) is an electrical machine that converts DC electrical energy into mechanical energy (rotation) using magnetic fields, fundamental to devices from toys to EVs, working by passing current through coils (rotor/armature) within a magnetic field (stator), creating force (Lorentz Force) that turns the rotor, with brushes and a commutator reversing current for continuous spin, offering excellent speed control and instant start/stop/reversal.

Figure 1.4: DC motor

- **<u>Lithium polymer battery and case(3.7v</u>**): A lithium polymer (LiPo) battery is a type of rechargeable battery that uses a solid or gel-like polymer electrolyte, enclosed in a flexible, soft aluminum foil pouch, which allows for versatile shapes and lightweight designs. A "case" for this type of battery can refer either to its integrated soft aluminum foil pouch or an external hard plastic holder used for mounting and protection in electronic project.

Figure 1.5: Lithium polymer battery

- **<u>Diod:</u>** A diode is a two-terminal electronic component, usually semiconductor-based, that acts like a one-way valve for electric current, allowing it to flow easily in one direction (forward bias) but blocking it from flowing in the opposite direction (reverse bias). It has an anode (positive) and a cathode (negative) and is crucial for converting AC to DC, protecting circuits, and controlling current flow in electronics.

Figure 1.6: Diod

- **<u>Mini solar panel:</u>** A mini solar panel is a small, compact photovoltaic (PV) device that converts sunlight into a limited amount of direct current (DC) electricity, typically generating 1 to 200 watts, ideal for charging small electronics, powering portable gadgets, or supplementing off-grid energy needs like RVs, camping lights, and security cameras, unlike large rooftop panels for homes.

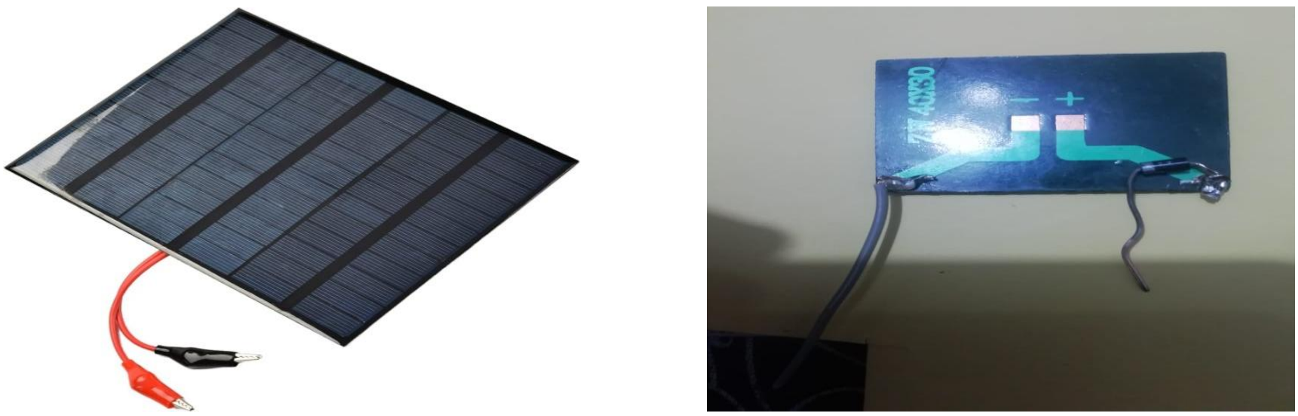

Figure 1.7: Mini solar panel

- **Wire:** A connecting wire is an electrical conductor, usually copper, used to link components in a circuit, allowing current to flow, available in various types (like silicone, PVC) and sizes (AWG), often insulated, and joined using connectors, twisting, or soldering for applications from simple school projects to complex electronics.

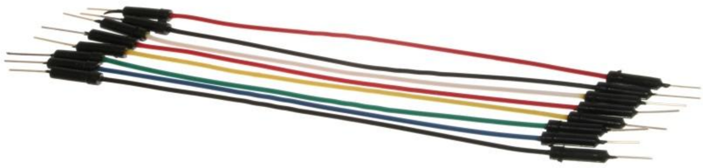

Figure 1.8: Wire

- **Push button switch**: A push-button switch is a simple electrical switch activated by a momentary push, opening or closing a circuit to control devices, returning to its original state (usually via a spring) when released, and is used everywhere from doorbells (momentary) to washing machines (latching) for on-demand control.

Figure 1.9: push button switch

❖ **Project description :** This project focuses on designing a sun-tracking solar panel system that automatically follows the sun to maximize energy generation. Using a LDR , a IC(TDA2822M) , and motor control, the panel continuously adjusts its position for optimal sunlight exposure, resulting in higher efficiency compared to fixed solar panels. The system reduces manual intervention and ensures effective utilization of solar energy throughout the day. It is suitable for renewable energy applications and demonstrates practical implementation of control and power electronics concepts. The project also highlights the importance of sustainable and eco-friendly energy solutions.

❖ Circuit Diagram:

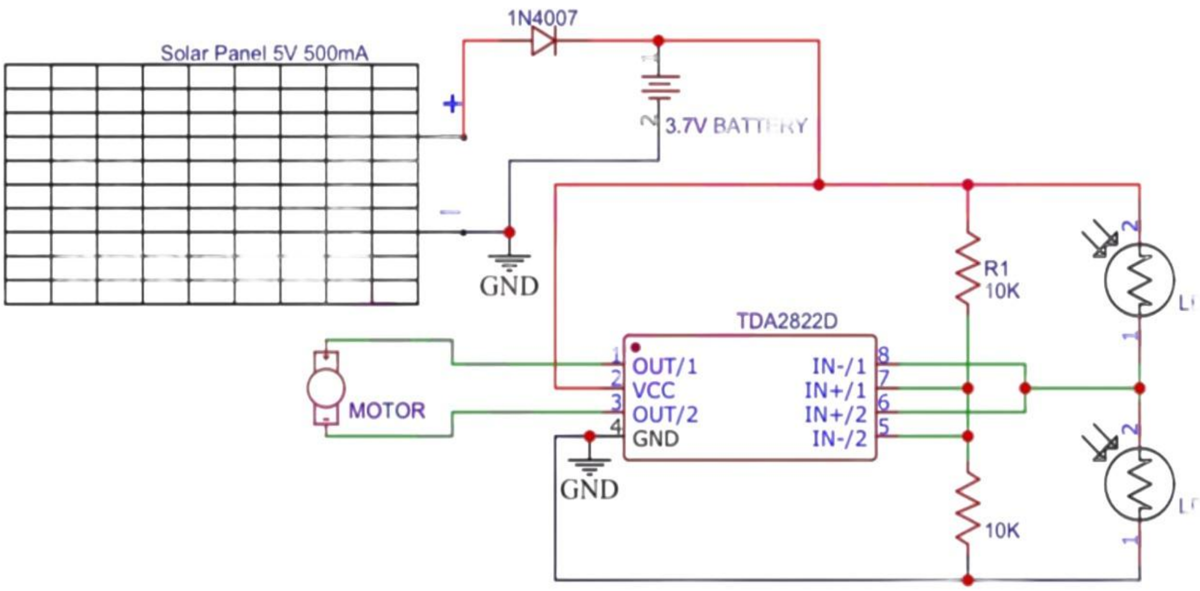

Figure 1.10: Sun tracking solar panel circuit diagram

❖ **Circuit Description :** The circuit consists of a 5 V solar panel that charges a 3.7 V battery through a 1N4007 diode, which prevents reverse current flow. Two LDRs are connected with 10 ohm resistors to form a voltage divider that senses light intensity from different directions. The voltage difference from the LDR network is fed to the TDA2822D IC, which is used as a driver to control the DC motor. Based on the light imbalance, the IC drives the motor to rotate the solar panel toward the brighter light source. When both LDRs receive equal light, the motor stops, keeping the panel aligned with the sun.

- **Working principle:** The circuit operates on the principle of light intensity comparison using LDRs. When sunlight falls unevenly on the two LDRs, a voltage difference is produced, which activates the motor driver IC to rotate the motor toward the brighter side. The solar panel continues to move until both LDRs receive equal light, ensuring the panel remains aligned with the sun for maximum energy generation.

- **Working Procedure :** Light Sensing (LDRs): Mount two LDRs (Light Dependent Resistors) on the panel, one on the left, one on the right. Connect each LDR with a fixed resistor (e.g., 220Ω or 10kΩ) to form two voltage dividers, creating varying voltages based on light intensity. Signal Processing (TDA2822): The TDA2822 IC contains op-amps that act as comparators. The outputs from the LDR voltage dividers are fed into the TDA2822's inputs. The chip compares these voltages to determine which side is brighter.
  Motor Control: If the left LDR is brighter (more light), the TDA2822 output drives the DC motor to rotate the panel left.
  If the right LDR is brighter, the motor rotates the panel right. If both are equal, the motor stops.
  Mechanical Movement: A small DC motor is mechanically linked to the solar panel. As the motor turns, it physically tilts the panel.
  Power & Protection: A battery (e.g., 3.7V) powers the circuit, often with a charging circuit for the panel. A diode (like 1N4007) protects the circuit from reverse current.

- **How we build and used the circuit :**

  - **Step 1: Building the Circuit and Sensor Setup:** All components such as the solar panel, LDRs, resistors, diode, battery, motor, motor driver IC, and wires were collected and tested. The LDRs were mounted on opposite sides of the solar panel to detect sunlight direction. Each LDR was connected with a resistor to form a voltage divider circuit, which produces different voltages based on light intensity.
  - **Step 2: Motor and Power Circuit Construction:** The output from the LDR voltage divider circuit was connected to the input of the motor driver IC, and a DC motor was connected to its output to rotate the panel. The solar panel was connected to a rechargeable battery through a diode to prevent reverse current flow. The battery supplies power to the motor driver and the overall circuit.
  - **Step 3: Using the Circuit Under Sunlight**: After completing the circuit, it was placed under sunlight for operation. When sunlight fell more on one LDR than the other, a voltage difference was created, activating the motor driver. The motor rotated the solar panel toward the direction of higher light intensity.
  - **Step 4: Automatic Tracking and Energy Generation:** The panel continued to rotate until both LDRs received equal sunlight, after which the motor stopped automatically. In this aligned position, the solar panel generated maximum power and charged the battery efficiently without any manual control.

❖ Hardware working snap short:

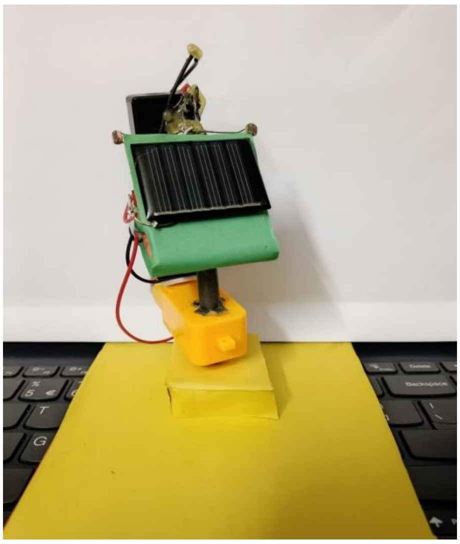

Figure 1.11: Hardware front view

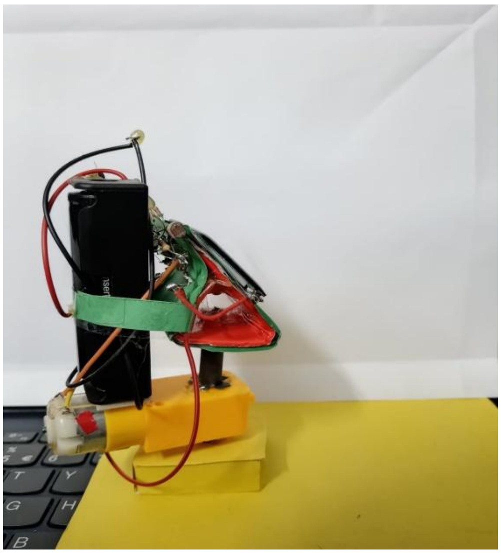

Figure 1.12: Hardware side view

❖ Key Notes:

1. **Sunlight Detection:** Light Dependent Resistors (LDRs) are used to sense sunlight from different directions.The difference in light intensity helps determine the direction of movement.

2. **Analog Control System:** The project is fully based on analog components without using any microcontroller or software. This makes the circuit simple, low-cost, and easy to understand.

3. **Motor Control Mechanism**: A motor driver IC controls the DC motor based on LDR signal variation. The motor rotates the solar panel toward the brighter light source.

4. **Automatic Panel Alignment:** The panel moves automatically until both LDRs receive equal light. At this point, the motor stops, keeping the panel aligned with the sun.

5. **Power Generation Unit:** A solar panel converts sunlight into electrical energy. The generated power is used to charge the battery and operate the circuit.

6. **Battery Charging and Protection:** A diode is used to prevent reverse current flow from the battery to the solar panel. This ensures safe charging and protects the solar panel.

7. **Energy Efficiency Improvement:** Continuous sun tracking increases energy output compared to fixed panels. This improves overall system efficiency and performance.

8. **Mechanical Structure:** A stable mounting frame supports the solar panel and motor assembly. It allows smooth and controlled rotation of the panel.

9. **Application and Use:** Suitable for renewable energy systems and educational EEE projects. Demonstrates practical use of sensors, power electronics, and control concepts.

- **<u>Discussion:</u>** The sun-tracking solar panel project demonstrates an effective method of improving solar energy efficiency using a simple and low-cost analog circuit. By continuously adjusting the panel position based on light intensity detected by LDRs, the system ensures maximum exposure to sunlight throughout the day. Unlike fixed solar panels, this automatic tracking significantly increases power generation without manual intervention. The absence of microcontrollers and software makes the project easy to construct, reliable, and suitable for educational purposes. Overall, the project highlights the practical application of renewable energy concepts and emphasizes the importance of sustainable power generation.